

Semantic Secrets: Private Authentication from Reconstructable Visual Semantics

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Abstract

This planning manuscript defines and tests a revised research direction; it does not report a completed authentication system. We ask whether a remembered visual concept can be reconstructed through independently generated images, converted locally into a canonical typed semantic graph, and privately verified under a system-derived policy. The prompt is a reconstruction interface rather than the secret, and generated pixels are transient observations rather than entropy. A focused P8 primary-source review found that generative graphical authentication, flexible semantic policy, fuzzy/private matching, protected templates, AI-assisted guessing, and adaptive verifier attacks are all prior art. Gate V2-N therefore passed only after narrowing the remaining question and freezing the graph, data isolation, model limits, attacks, uncertainty, and numeric gates. P9 then tested two frozen local extractors on a new controlled capability set. Each produced an invalid graph on its first formal validation fixture. With 32 validation fixtures, one invalid output caps best-case schema validity at $31/32 = 96.875\%$, below the preregistered 98% requirement. Neither extractor advanced, P9B was not executed, Gate V2-A failed, and policy and cryptographic phases remain blocked. This bounded negative result rejects the tested representation path, not every possible visual-semantic extractor.

1 Problem and scope

Authentication based on approximate semantic meaning has an unusual tension. Benign reconstructions must vary, yet every tolerated variation enlarges the set an attacker may submit. If a server stores a deterministic semantic verifier, the same structure that permits fuzzy acceptance may reveal semantic content, support offline candidate testing, or link reuse across services. Secure computation may hide the comparison while leaving the accepted region easily enumerable.

We study a prospective client-side pipeline in which a person reconstructs a remembered visual concept using free natural language. A local generator produces a new image, and a local extractor maps that image into a canonical semantic representation. The strongest target sends neither prompt, image, plaintext graph, nor raw embedding to the authentication infrastructure.

No participant study is planned. Researcher-authored concepts and reconstructions can measure only technical semantic reconstruction stability. They cannot establish memorability, recall, usability, preference, or natural credential choice.

2 The prompt and image are not the secret

Let C denote a remembered visual concept and P a free-form reconstruction prompt. Different expressions may aim at the same concept—for example, “a cat beside a red car” and “a feline next to

a crimson automobile.” The hypothesis is not exact prompt reproduction. With generator G and randomness r ,

$$I = G(P, r), \quad (1)$$

where I is a transient, independently generated image. Pixels do not contribute password entropy merely because generation is stochastic. A versioned extractor E produces

$$S = E(I). \quad (2)$$

The credential source is the canonical semantics and, operationally, the complete set of candidate semantics the verifier accepts.

This framing differs from treating direct text embeddings or a single generated image as the authentication object. It also creates a falsifiable condition: if independent generation and fresh extraction do not preserve security-relevant structure, the visual reconstruction hypothesis fails.

3 Canonical policy-aware semantics

We provisionally represent S as a typed graph

$$S = (V, E_S, A, C_S), \quad (3)$$

where V contains entities, E_S typed actions/relations, A attributes, and C_S counts and scene/context facts. A deterministic, versioned policy derivation function produces

$$\Pi(S) = (M, T), \quad (4)$$

where M contains mandatory security anchors and T contains secondary facts with bounded tolerance. The primary non-human study will not ask users to nominate M or T . Selection must be frozen from independent semantic rationale, training-only statistics, extractor reliability, attacker-frequency evidence, and protocol compatibility.

A P8-frozen reference functionality has the form

$$\text{Accept}_{\text{ref}} = \text{MandatoryOK}(M, M') \wedge [\text{TolSim}(T, T') \geq \tau], \quad (5)$$

The exact closed vocabulary, canonical graph correspondence, policy eligibility and ranking, tolerant weighted-F1 semantics, malformed/version behavior, threshold grid and development-only selector are specified in docs/formal_specification_v2.md. The notation and policy split are not claimed as standalone novelty.

4 Acceptance-region security

For an enrolled state, define the accepted region

$$\mathcal{A}(S, M, T) = \{(S', M', T') : \text{Accept}_{\text{ref}} = 1\}. \quad (6)$$

An attacker succeeds by finding any member, not by recovering the enrolled prompt, image, or exact graph. We distinguish:

- K_0 : generic public knowledge and random candidates;
- K_1 : population or frequency knowledge without target facts;
- K_2 : partial target-specific semantic information;
- K_3 : strong near-secret information.

For strategy Q , budget B , knowledge K_i , and observable view V , the primary quantity is

$$\text{Succ}(Q, B, K_i, V) = \Pr[\exists j \leq B : q_j \in \mathcal{A}(S, M, T)]. \quad (7)$$

The evaluation should report success@1, @5, @10, and @B, guesses to success, duplicate policy, uncertainty, and cost. Random FAR is only one distributional statistic. Frequency, targeted neighbours, partial knowledge, LLM/VLM/generator assistance, and adaptive semantic search may produce different risk. AI-assisted password guessing is established prior art [18]; this project must contribute a semantic acceptance-region analysis rather than the general idea of using a model to guess.

5 Privacy target

The protected record is abstractly

$$R = \text{Protect}(S, M, T, \text{account}, \text{domain}, \text{version}). \quad (8)$$

A later protocol should reproduce the frozen plaintext decision while revealing no more than its declared output and metadata. Candidate goals are:

- template confidentiality against semantic recovery or accepted-equivalent reconstruction;
- resistance to cheap database-only candidate validation under an exact stolen-record view;
- policy privacy for anchor identities, operators, weights, thresholds, and graph shape;
- unlinkable, domain-separated records and transcripts;
- transcript minimisation, replay/session binding, and explicit adaptive-query leakage;
- containment under named database, server, key/share, collusion, and total-compromise states.

Established fuzzy extractors [4], fuzzy aPAKE [5], OPRFs [3], OPAQUE [2], exact PSI [6, 17], fuzzy PSI/private biometric matching [1, 20], committed PSI [7], private fuzzy-record computation [9], private graph operations [11], and homomorphic vector matching [12] are candidate foundations, not novel primitives. Raw learned representations are not automatically private; embedding leakage is an explicit concern [19]. Protected-template unlinkability and irreversibility must be attacked rather than inferred from an encoding [8].

Semantic correctness and cryptographic privacy are independent. FAR/FRR do not prove privacy; private computation does not repair an easily guessed acceptance region.

6 Closest work and VSA boundary

VSA [10] is the closest mandatory baseline. It enrolls a reference image, extracts objects, CLIP-derived attributes, quantities, visibility, and 2x2 quadrant location, and lets a user select two or three semantic facts plus quantity operators. Authentication uses an arbitrary satisfying image and an additional password. VSA serialises canonical semantic tokens, evaluates Flexible Range Logic conjunctions, hashes most semantic fields together with the password using SHA-256, and retains quantity/operator metadata separately.

These features remove image-independent visual-semantic authentication, VLM extraction, canonical tokens, object, attribute, count, and quadrant rules, flexible quantity policies, conjunction,

and user-selected semantic policy from our novelty space. We require two fair baselines: (i) VSA’s semantic-policy component for functional discrimination and (ii) the full semantic-plus-password hash architecture for security comparison. We must not compare a semantic-only system directly with VSA’s claimed combined semantic-plus-password space as though the credentials were equivalent.

VSA calls its hash binding server-opaque and explicitly weaker than strict zero knowledge. It acknowledges small semantic inputs and fast SHA-256 offline guessing, recommending Argon2id with a user-specific salt. Its COCO-based positive FRR samples are automatically selected from cached detections rather than independently re-inferred. It reports no human study and leaves Semantic Relationship Logic to future work. It does not target private/unlinkable verification.

SemanticLock uses semantically linked image tokens and story structure for graphical authentication [16]. PassStyles [14] and Omokage [15] use StyleGAN-generated changing face candidates and recognition criteria, preventing broad “first semantic”, “first generative”, or “first changing-image” authentication language. Adaptive score-guided verifier search is likewise established [13]. The P8 review found no verified source combining free-language independent text-to-image reconstruction and fresh typed extraction with a system-derived policy, finite-budget semantic attacks, and construction-specific private/domain-separated verification. Gate V2-N passed for that scoped question only; this is not an exhaustive absence or first claim.

7 Research questions

RQ1—Reconstruction. Can independently generated images representing the same intended visual concept produce stable canonical semantic credentials?

RQ2—Policy-aware matching. Can a system-derived $S = (M, T)$ policy preserve legitimate technical reconstruction variation while rejecting security-critical semantic differences?

RQ3—Acceptance-region security. How vulnerable is the induced accepted region under K_0 – K_3 attackers and finite budgets?

RQ4—Template/privacy leakage. What semantics can be inferred from stored records, policies, and authentication transcripts under named compromise views?

RQ5—Private verification. Can the frozen policy be evaluated while hiding the semantic credential and resisting practical database-only validation and cross-domain linkage under stated assumptions?

RQ6—Practicality. What computation, bandwidth, latency, storage, trust, and deployment assumptions are required?

8 Proposed methodology and baselines

The execution order is deliberately staged.

Novelty and preregistration. P8 completed a focused primary-source novelty refresh and froze graph tasks, controlled concepts, splits, capability screening, policy derivation, attack budgets, uncertainty, and numeric gate thresholds. It performed no model acquisition, dataset acquisition, v2 inference, sealed-v1 access, or cryptographic implementation.

Representation before authentication. P9A screened two preregistered extractors on 96 new controlled images with known object, attribute, count, action, relation, and context ground truth. Both failed the schema-validity prerequisite under exact logical futility. P9B was restricted to P9A survivors and was therefore not executed. Direct text remained an ablation rather than the target credential path.

Policy and attackers. P10 compares dense similarity, global fuzzy overlap, a feasible VSA-style policy, and the proposed system-derived M/T policy. P11 evaluates the full acceptance region under budgeted attackers before any cryptographic implementation.

Private verification. Only a constructive semantic-security gate permits P12 to define the ideal functionality and compare mature constructions. Minimal POCs precede selection. The selected protocol is then evaluated for reference correctness, leakage, offline validation, compromise, linkage, and cost.

Baselines. Planned baselines include direct-text processing, dense visual/text similarity, global fuzzy semantic overlap, VSA semantic-policy behaviour, the full VSA binding model where reproducible, plaintext/hash records, and relevant mature private-matching constructions. Any inaccessible or underspecified VSA behaviour will be documented rather than guessed.

9 P9 capability outcome

The P9 capability manifest contains 64 development and 32 validation fixtures, equally divided between procedural/composite and researcher-authored synthetic-raster scenes. It accesses no participant or sealed-v1 data. Moondream2 at revision prefix 9a7d40240508 emitted no JSON on formal validation fixture cap-v2-064 in 157.57 seconds with 4.58 GiB peak process RSS. SmolVLM2-2.2B-Instruct at revision prefix c0a7af506d0f emitted malformed, truncated JSON in 392.10 seconds with 5.62 GiB peak RSS. The artifact records both complete immutable revisions.

The preregistration did not specify an early-stop rule. After separate plumbing smoke and before formal validation output, we implemented exact logical futility: because all checks are conjunctive, one invalid result makes the 98% schema-validity check impossible even if all remaining fixtures pass. This procedural deviation cannot create a pass, but it limits the negative evidence. We did not estimate full atom F1, bootstrap intervals, determinism, structured error strata, or full-set latency medians. We made no candidate replacement, prompt/output repair, fine-tuning, or authentication-outcome selection.

10 Decision gates

The gates are: V2-N novelty viability; V2-A visual-semantic reconstruction viability; V2-B policy-aware semantic viability; V2-C acceptance-region security viability; V2-D private-functionality viability; V2-E protocol selection/privacy viability; and V2-F end-to-end evidence/paper story. No later cost may bypass an earlier failure. Valid outcomes include standalone authentication, second factor, policy-constrained credential, measurement/negative contribution, or stop.

11 Formative v1 evidence

The completed P0–P7 programme evaluated a different system, visual-semantic-pipeline-v1. P6’s training-selected weighted matcher had validation FRR 20.8%, targeted-neighbour FAR 75.0%, and random FAR 3.0%; all tested representations failed the original gate. Twelve v1 test families remain sealed. P7 then found no uncertainty-supported material benefit from the tested SD-Turbo/Florence/SigLLIP image pathways and removed that image stage from the v1 authentication core.

These are immutable formative negative results, not v2 measurements. They motivate explicit security anchors and independent capability screening but may not tune the v2 policy, representation, data, or gates. The v2 hypothesis requires new identifiers, preregistration, and evidence.

12 Expected contributions and limitations

If all gates pass, C1 would establish technical behaviour of the scoped free-language generative reconstruction path; C2 would establish only the measured incremental value of the frozen system-derived typed policy; C3 would characterise the accepted region under finite-budget attackers; and C4 would establish construction-specific private and unlinkable verification. C3/C4 carry the prospective principal contribution. P8 rejected C5 as standalone and retained adaptive semantic verification-oracle leakage as mandatory evidence inside C3/C4.

The frozen path stopped at the first risk: neither tested extractor met the structured-output prerequisite. Whether another independently justified representation could work remains unresolved, as do the downstream policy, guessing-region, and private-construction questions; they were not authorised after Gate V2-A failed. The controlled methodology cannot establish human memorability or usability. A future participant study would require separate ethics, design, and evidence.

13 Conclusion

This work treats visual generation as a testable reconstruction layer, not a source of magical entropy, and privacy as a construction-specific property rather than a consequence of hashing semantics. P8 passed Gate V2-N with narrowed claims, but P9 falsified the tested frozen extractor path at its first independent capability prerequisite. P9B, policy optimisation, attacker evaluation, and cryptographic prototyping were correctly withheld. The project must now stop or explicitly reframe with a new representation, rationale, data, preregistration, and gate; the failed P9 evidence cannot be tuned into a pass.

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